

CASE 01: The Cozy Coffee Corner Company

A Python-Powered Coffee Shop Analytics Challenge

Dr. Tim Smith

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The Business Challenge

Company Background: The Cozy Coffee Corner

The Story

The Cozy Coffee Corner is a thriving independent coffee shop located in Tampa's bustling Westshore business district. What started as a single location three years ago has grown into a local favorite, known for artisanal coffee, fresh pastries, and a warm atmosphere that keeps customers coming back.

The Current Situation

Sarah Martinez, the owner and founder, has built her business on intuition and personal relationships with customers. However, as the business has grown, she's facing new challenges:

- **Growing Competition:** Two national chains have opened nearby in the past six months
- **Changing Customer Patterns:** Remote work has shifted peak hours and customer preferences
- **Inventory Management:** Some items sell out early while others go to waste
- **Loyalty Program Questions:** Unsure if the membership program is driving enough value
- **Expansion Decision:** Considering opening a second location but needs data to support the investment

The Data Opportunity

Sarah has been diligently collecting data through her point-of-sale system, but it's scattered across spreadsheets and she lacks the technical skills to analyze it effectively. She's heard about your Python expertise and wants your help to transform this data into actionable insights.

Your Mission

Use Python and the concepts from Weeks 1-3 to help Sarah understand her business better and make data-driven decisions about inventory, customer loyalty, and potential expansion.

The Available Data

Data Collection Overview

Sarah has provided you with several months of data, but we'll start with analyzing a typical week to establish patterns. The data includes:

1. Menu and Pricing Information

```
# Current menu with prices (as of January 2025)
menu_prices = {
    "Espresso": 2.50,
    "Americano": 3.00,
    "Latte": 4.50,
    "Cappuccino": 4.00,
    "Macchiato": 4.75,
    "Mocha": 5.00,
    "Cold Brew": 3.50,
    "Iced Latte": 4.75,
    "Croissant": 3.50,
    "Blueberry Muffin": 3.00,
    "Bagel & Cream Cheese": 4.00,
    "Avocado Toast": 8.50,
    "Breakfast Sandwich": 7.50,
    "Cookie": 2.00,
    "Brownie": 3.50
}

# Product categories (using tuples for immutable groupings)
beverages = ("Espresso", "Americano", "Latte", "Cappuccino",
            "Macchiato", "Mocha", "Cold Brew", "Iced Latte")
food_items = ("Croissant", "Blueberry Muffin", "Bagel & Cream Cheese",
            "Avocado Toast", "Breakfast Sandwich", "Cookie", "Brownie")

# Store hours as a tuple (immutable schedule)
store_hours = ("Monday", "6:00 AM", "8:00 PM"),
              ("Tuesday", "6:00 AM", "8:00 PM"),
              ("Wednesday", "6:00 AM", "8:00 PM"),
              ("Thursday", "6:00 AM", "9:00 PM"),
              ("Friday", "6:00 AM", "9:00 PM"),
```

```
("Saturday", "7:00 AM", "9:00 PM"),
("Sunday", "8:00 AM", "6:00 PM"))
```

2. Customer Database

```
# Customer information (list of dictionaries)
customers = [
    {"id": "C001", "name": "Alice Johnson", "member": True, "join_date": "2024-01-15"},
    {"id": "C002", "name": "Bob Smith", "member": False, "join_date": None, "status": "inactive"},
    {"id": "C003", "name": "Charlie Davis", "member": True, "join_date": "2024-03-01"},
    {"id": "C004", "name": "Diana Wilson", "member": True, "join_date": "2023-11-20"},
    {"id": "C005", "name": "Eric Brown", "member": False, "join_date": None, "status": "inactive"},
    {"id": "C006", "name": "Fiona Taylor", "member": True, "join_date": "2024-02-10"},
    {"id": "C007", "name": "George Martinez", "member": False, "join_date": None, "status": "inactive"},
    {"id": "C008", "name": "Helen Anderson", "member": True, "join_date": "2022-09-05"},
    {"id": "C009", "name": "Ivan Rodriguez", "member": True, "join_date": "2024-04-01"},
    {"id": "C010", "name": "Julia Chen", "member": False, "join_date": None, "status": "inactive"}
]

# Loyalty program benefits (stored as a set for unique perks)
member_benefits = {"10% discount", "free birthday drink", "early access to new menu items",
                   "double points on Wednesdays", "free size upgrade"}
```

3. Daily Sales Data

```
# Monday's orders (each tuple contains: customer_id, items ordered, time)
monday_orders = [
    ("C001", ["Latte", "Croissant"], "7:15 AM"),
    ("C003", ["Americano", "Americano", "Bagel & Cream Cheese", "Brownie"], "7:00 AM"),
    ("C008", ["Cappuccino"], "8:00 AM"),
    ("C002", ["Espresso", "Cookie"], "8:15 AM"),
    ("C004", ["Mocha", "Avocado Toast"], "8:30 AM"),
    ("C007", ["Latte", "Latte", "Blueberry Muffin"], "9:00 AM"),
    ("C001", ["Iced Latte"], "12:30 PM"),
    ("C006", ["Cold Brew", "Breakfast Sandwich"], "1:00 PM"),
    ("C009", ["Americano", "Cookie", "Cookie"], "2:15 PM"),
    ("C004", ["Latte"], "3:30 PM"),
    ("C010", ["Espresso"], "4:00 PM"),
]
```

```

    ("C003", ["Cappuccino", "Brownie"], "5:00 PM")
]

# Tuesday's orders
tuesday_orders = [
    ("C001", ["Latte", "Breakfast Sandwich"], "7:00 AM"),
    ("C008", ["Americano", "Croissant", "Croissant"], "7:20 AM"),
    ("C004", ["Mocha"], "8:00 AM"),
    ("C005", ["Cold Brew", "Bagel & Cream Cheese"], "8:45 AM"),
    ("C006", ["Cappuccino", "Blueberry Muffin"], "9:15 AM"),
    ("C001", ["Espresso"], "11:00 AM"),
    ("C003", ["Iced Latte", "Avocado Toast"], "12:00 PM"),
    ("C009", ["Latte", "Cookie"], "2:00 PM"),
    ("C002", ["Americano"], "3:15 PM"),
    ("C004", ["Cold Brew", "Brownie"], "4:30 PM"),
    ("C007", ["Mocha", "Mocha", "Breakfast Sandwich"], "5:15 PM"),
    ("C008", ["Cappuccino"], "6:00 PM")
]

# Competitor information (for market analysis)
nearby_competitors = {
    "Starbucks": {"distance": "0.3 miles", "avg_coffee_price": 5.50},
    "Dunkin": {"distance": "0.5 miles", "avg_coffee_price": 3.50},
    "Local Bean": {"distance": "0.8 miles", "avg_coffee_price": 4.00}
}

```

Business Questions to Solve

Core Analytics Challenges

Using Python and the concepts you've learned in Weeks 1-3, help Sarah answer these critical business questions:

1. Sales Performance Analysis

Questions to Answer

1. What is the total revenue for Monday and Tuesday?
2. What is the average order value (AOV) for each day?
3. Which items are the best sellers by quantity and by revenue?
4. What percentage of revenue comes from food vs. beverages?

2. Customer Behavior Insights

Questions to Answer

1. What percentage of customers are loyalty members?
2. Do members spend more on average than non-members?
3. Who are the top 3 customers by visit frequency?
4. What are the peak ordering times?

3. Competitive Positioning

Questions to Answer

1. How do our prices compare to competitors?
2. Which unique items differentiate us from chains?
3. What's our price advantage/disadvantage for common items?

4. Inventory Optimization

Questions to Answer

1. Which items haven't sold at all?
2. What items appear in orders together frequently (basket analysis)?
3. Should we adjust our menu based on sales patterns?

Your Task: Python-Powered Solutions

Implementation Requirements

Phase 1: Data Processing with Functions

Create reusable functions to process the data:

```
def calculate_order_total(order_items, prices, is_member=False):
    """
    Calculate total for an order, applying member discount if applicable

    Parameters:
    - order_items: list of item names
    - prices: dictionary of item prices
    - is_member: boolean indicating membership status

    Returns:
    - float: total order amount
    """
    # Your implementation here
    pass

def analyze_sales_by_category(orders, prices, beverages, food_items):
    """
    Separate sales into beverage and food categories

    Returns:
    - tuple: (beverage_revenue, food_revenue)
    """
    # Your implementation here
    pass

def find_popular_combinations(orders):
    """
    Identify items frequently ordered together

    Returns:
    - set: unique combinations of items
    """
    # Your implementation here
    pass
```

Phase 2: Customer Analytics

Analyze customer patterns using the data structures:

```
def segment_customers(customers, orders):  
    """  
    Segment customers into categories based on behavior  
  
    Returns:  
    - dictionary: customer segments with their characteristics  
    """  
    # Your implementation here  
    pass  
  
def calculate_member_value(customers, orders, prices):  
    """  
    Compare member vs non-member spending patterns  
  
    Returns:  
    - dictionary: statistics for each group  
    """  
    # Your implementation here  
    pass
```

Phase 3: Strategic Recommendations

Based on your analysis, provide Sarah with:

1. **Data-driven insights** about her business
 2. **Specific recommendations** for improvement
 3. **Metrics to track** going forward
 4. **Decision support** for the expansion question
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Expected Deliverables

What Sarah Needs From You

1. Executive Summary Dashboard

Create a clear output showing: - Daily and weekly revenue totals - Top 5 products by revenue - Customer segmentation results - Key performance indicators (KPIs)

2. Detailed Analysis Report

Provide insights on: - Sales patterns and trends - Customer behavior analysis - Product performance metrics - Competitive positioning

3. Actionable Recommendations

Suggest specific actions such as: - Menu optimization strategies - Loyalty program improvements - Pricing adjustments - Inventory management changes - Expansion feasibility assessment

4. Python Implementation

Your code should demonstrate: - **Functions:** Modular, reusable code - **Data Structures:** Appropriate use of lists, dictionaries, tuples, and sets - **Control Flow:** Loops and conditionals for data processing - **Best Practices:** Clear variable names, comments, and organization

Hints and Guidance

Getting Started

Approach Strategy

1. Start by understanding the data structure
2. Write simple functions to calculate basic metrics
3. Build more complex analyses on top of basic functions
4. Test with small data samples before processing everything

Key Concepts to Apply

From Week 1: Foundations

- Variables and data types
- Basic operations and calculations
- String formatting for output

From Week 2: Control Flow & Data Structures

- `for` loops to iterate through orders
- `if/elif/else` for conditional logic
- Lists for ordered data
- Dictionaries for key-value mappings

From Week 3: Advanced Structures & Functions

- Tuples for immutable data (like store hours)
- Sets for unique collections (like benefits)
- Functions for code reusability
- Return values and parameters

Common Pitfalls to Avoid



Watch Out For

- Don't forget to handle member discounts in calculations
- Check if items exist in the price dictionary before accessing
- Consider edge cases (empty orders, new customers)
- Remember that sets remove duplicates automatically

Extension Challenges

For Advanced Students

If you complete the main requirements, consider these additional challenges:

Challenge 1: Time-Based Analysis

Create a function that analyzes sales patterns by time of day to identify rush hours and slow periods.

Challenge 2: Customer Lifetime Value

Calculate the projected lifetime value of members vs. non-members based on visit frequency and average spending.

Challenge 3: Menu Engineering

Use the concept of menu engineering (popularity vs. profitability) to categorize items into: - Stars (high popularity, high profit) - Plowhorses (high popularity, low profit) - Puzzles (low popularity, high profit) - Dogs (low popularity, low profit)

Challenge 4: Predictive Insights

Based on the patterns you find, predict: - Which customers are likely to become members - What tomorrow's sales might look like - Which items should be promoted together

Conclusion

Making Data-Driven Decisions

This case study represents a real-world scenario where Python programming skills can transform raw business data into actionable insights. By completing this analysis, you'll help Sarah:

1. **Understand** her business performance objectively
2. **Identify** opportunities for growth and optimization
3. **Make** informed decisions about expansion
4. **Compete** effectively against larger chains

! Remember

The goal isn't just to write code that works—it's to provide insights that can genuinely help Sarah's business thrive. Think like a business analyst who happens to know Python!

Good Luck!

Sarah is counting on your analytical skills to help The Cozy Coffee Corner continue to grow and serve the community. Your Python-powered insights could be the key to the shop's continued success!

This case study integrates concepts from ISM4641 Weeks 1-3. For questions or clarifications, please reach out to Dr. Smith or the class TA.